BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023 • Paper Code: 8483

Retyped faithfully from the original board paper — same questions, no changes.

Chemistry (Objective) Group - I

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8483

Note: Q.No.1. You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. The main water pollutant chromium-VI is discharged by:

(A) Plastic industry
(B) Paper industry
(C) Leather industry
(D) Cement industry

Q2. Urea contains:

(A) 36% nitrogen
(B) 46% nitrogen
(C) 56% nitrogen
(D) 66% nitrogen

Q3. Polyamide resins are:

(A) Homopolymer
(B) Copolymer
(C) Terpolymer
(D) Addition polymer

Q4. Compounds having - C ≡ N group are called:

(A) Nitro compound
(B) Amino acid
(C) Alkane nitriles
(D) Amide

Q5. During reduction of aldehyde with $NaBH_4$, H^- ion act as:

(A) Electrophile
(B) Nucleophile
(C) Acid
(D) Base

Q6. Which enzyme is not involved in fermentation of starch?

(A) Diastase
(B) Zymase
(C) Urease
(D) Maltase

Q7. Carbolic acid has another name of:

(A) Alcohol
(B) Phenol
(C) Ether
(D) Carboxylic acid

Q8. Which is not a nucleophile?

(A) H_2O
(B) H_2S
(C) BF_3
(D) NH_3

Q9. The electrophile in aromatic sulphonation is:

(A) H_2SO_4
(B) HSO_4^{-1}
(C) SO_3
(D) SO_3^{+}

Q10. β-β'-dichloroethyl sulphide is known as:

(A) Mustard gas
(B) Laughing gas
(C) Phosgene gas
(D) Bio gas

Q11. A double bond consists of:

(A) Two sigma bonds
(B) One sigma and one pi bond
(C) One sigma and two pi bond
(D) Two pi bond

Q12. The colour of transition metal complexes is due to:

(A) d-d transition of electrons
(B) Ionization
(C) Loss of s-electron
(D) Gain of s-electron

Q13. Bleaching powder is prepared by passing chlorine over:

(A) Calcium carbonate
(B) Calcium sulphate
(C) Calcium hydroxide
(D) Magnesium hydroxide

Q14. Catalyst used in contact process is:

(A) Fe_2O_3
(B) V_2O_5
(C) SO_3
(D) Ag_2O

Q15. The chief ore of aluminium is:

(A) Na_3AlF_6
(B) $Al_2O_3.2H_2O$
(C) Al_2O_3
(D) $Al_2O_3.H_2O$

Q16. The mineral $CaSO_4.2H_2O$ has the general name:

(A) Gypsum
(B) Dolomite
(C) Calcite
(D) Epsom salt

Q17. Melting point of halogens:

(A) Decrease down the group
(B) Increase down the group
(C) Remains same in group
(D) First increase and then down the group

Chemistry (Subjective) Group - I

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 8483

> Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) How will you convert boric acid into borax and vice versa?

(ii) Write the formula of these minerals: (a) borax (b) Colemanite

(iii) What are the common properties of group IVA elements? (any four)

(iv) What is Wurtz-fitting reaction?

(v) How is glyoxal produced from benzene?

(vi) Define aromatic compounds with two examples.

(vii) What is condensation polymerization?

(viii) How can you differentiate between glucose and fructose?

(ix) Write any four uses of lipids?

(x) What is dissolved oxygen?

(xi) How do CO_2 and SO_2 cause acid rain?

(xii) How do pesticides affect living organism?

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) Write the functional group of alkanone and alkanoic acid with one example in each case.

(ii) What do you know about functional group isomerism? Give one example.

(iii) Give structural formula of alkene expected to be formed by dehydrohalogenation of 1-chloropentane.

(iv) Starting from ethene prepare ethyne.

(v) How is acetylene converted into chloroprene?

(vi) How does Grignard reagent react with methanal?

(vii) Starting from ethyl chloride prepare (a) n-butane (b) ethane.

(viii) Write the names of woody raw materials of paper industry.

(ix) What are the four essential qualities of a good fertilizer?

(x) Write the allotropic forms of phosphorus.

(xi) Write reaction taking place in contact tower for the manufacturing sulphuric acid.

(xii) Write the ring test for the confirmation of nitrate ion in solution.

Q4. (Marks 12) Write short answers to any SIX parts.

(i) What are ligands? Give one example

(ii) Draw the geometry of PCl_5

(iii) What is chromyl chloride test?

(iv) Define fermentation. Give one example.

(v) What is Lucas Test?

(vi) Why phenol is acidic but alcohol is not?

(vii) Give mechanism of addition of NH_3 with acetone.

(viii) What is iodoform test? Give its use.

(ix) What are amino acids? Give their general formula.

> Note: SECTION - II

> Note: Attempt any THREE questions. Each question carries 08 marks.

Q5.

(a) [08] Discuss the position of hydrogen with group I-A elements.

(b) [08] Write a detailed note on the commercial preparation of sodium by Down's cell.

Q6.

(a) [08] Write eight applications of noble gases.

(b) [08] Describe the process of digestion in paper industry.

Q7.

(a) [08] Explain any four features of organic compounds.

(b) [08] Explain the structure of benzene on the basis of molecular orbital treatment.

Q8.

(a) [02] Convert: (i) Ethyne into oxalic acid (ii) Propyne into acetone

(b) [01] What are alkyl halides? How alkyl halides are prepared from alcohol by three different reactions.

Q9.

(a) [01,01,01] How does acetaldehyde react with the following reagents? (i) C_2H_5MgI (ii) HCN (iii) $NaHSO_3$ (iv) dil NaOH

(b) [08] Discuss two methods of preparation of α-amino acids.